

CodeSpec AI — Engineering Report

Day 2 Implementation Summary: Architecture & Operational Dashboard

Project: CodeSpec AI	Phase: Day 2 (Dashboard)	Design System: Obsidian Sage
Platform: React 19 + Vite + Zustand	Backend API: FastAPI (/api)	Status: COMPLETED & VERIFIED

1. Executive Overview & Scope

Day 2 focused strictly on implementing the **Dashboard** interface on top of the established Day 1 foundation. The Dashboard functions as the central operational cockpit for CodeSpec AI, delivering immediate codebase awareness, AST-derived structural metrics, a high-level system topology preview, real-time repository change logs, and proactive documentation drift alerts without introducing unnecessary redesigns or speculative APIs.

2. Obsidian Sage Design System Tokens

All visual tokens were strictly synchronized with the immutable **Obsidian Sage** enterprise palette in `variables.css` and `global.css`:

Category	Token Variable	HEX Value	Semantic Role
App Shell	--app-background	#121312	Primary Dark Canvas
Sidebar / Topbar	--sidebar / --topbar	#171916 / #151714	Navigation chrome (205px / 56px)
Cards	--card-background	#1D201B (Border: #363A32)	Structured content cards (7px radius)
Graph Canvas	--graph-background	#171A17 (Grid: #282C26)	Architecture graph visualization
Accent / Brand	--primary / --secondary	#A8B39A / #9C927B	Sage primary & muted gold accents
Graph Tiers	--graph-frontend / service	#9BA8B0 / #91A78A	Tier 1 Web UI / Tier 2 Microservices
Storage & External	--graph-database / cache / ext	#A49A82 / #B87870 / #8F9A8C	PostgreSQL, Redis cache, GitHub VCS

3. Dashboard Component Breakdown

Component	File Path	Description & Visual Responsibility
RepositoryCard	.../dashboard/RepositoryCard.jsx	Compact contextual header highlighting active repo (CodeSpec-Core-Engine), status pill (Analyzed), branch (main), commit hash (8f4a21d), language stack, and AST indexing time.
MetricCard Row	.../dashboard/MetricCard.jsx	5-column metric strip displaying Files (1,248), Services (12), APIs (48), Functions (2,340), and Dependencies (36) with Lucide icons and context badges.
ArchitecturePreview	.../dashboard/ArchitecturePreview.jsx	Primary visual anchor displaying high-level system topology across 4 tiers (Frontend, Services, Storage, External) on an SVG grid canvas with interactive node hover, edge flows, and tier legend.
RecentChanges	.../dashboard/RecentChanges.jsx	Compact right-column event stream tracking AST and source edits (e.g. auth validator, API router, Redis config, AST dependency parser) with author and relative time tags.
DocumentationAlerts	.../dashboard/DocumentationAlerts.jsx	Proactive drift detection surfacing missing OpenAPI specs, undocumented core functions, and outdated module signatures with calibrated severity colors (warning/danger).
Dashboard Page	pages/Dashboard.jsx	Layout composer assembling Repository Context → Metrics Strip → 2-Column Grid. Clean separation of concerns with no bloated inline logic.

4. Common Primitives & State Infrastructure

- **Common UI Kit:** Standardized reusable primitives in `src/components/common/` including Card, Badge, Button, Loading, EmptyState, and Modal.
- **Zustand Stores:** Established `repositoryStore.js`, `architectureStore.js`, and `uiStore.js` to manage reactive state without global pollution.
- **API Service Abstraction:** Configured centralized Axios instance in `api.js` with clean endpoints in `repositoryApi.js` and `architectureApi.js` adhering strictly to backend contracts.

5. Verification & Compliance Matrix

Check Item	Standard / Requirement	Verification Result
Design Fidelity	Obsidian Sage dark theme, no neon, no gradients	PASSED (100% compliant)
No Extra Overview	No redundant Repository Overview section	PASSED (Maximized preview space)
No Large Page Title	No oversized Dashboard / Overview titles	PASSED (Vertical efficiency achieved)
ESLint Analysis	Code cleanliness, unused variables check	PASSED (0 errors, 0 warnings)
Vite Production Build	Optimized bundle compilation	PASSED (277 kB JS, 7.78 kB CSS)
Browser Rendering	Layout responsiveness & routing integrity	PASSED (Zero overflow, fluid grid)